Features & matching

Same place. Different pixels.



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HOW CAN A SYSTEM FIND THE SAME BOOK AGAIN?

120 minutes · 2 × 5-minute activities · Visual reasoning first

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From edges to corresponding points

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Week 3

Smooth → gradients → edges

The missing question

Which exact point is this?

Week 4

Locate → describe → match → verify

Week 3 measured local change. Week 4 uses it to find the same place again.

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Four goals you can see

LOCATE



A point is marked at a repeatable image location.

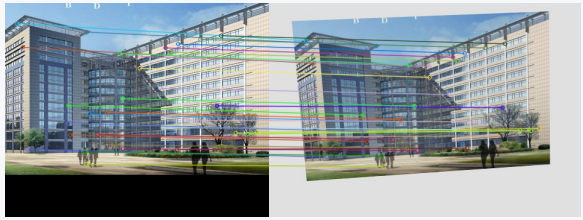
DESCRIBE



→ **[18, 0, 5, …]**
local appearance

The surrounding patch becomes a vector of numbers.

VERIFY



Keep pairs supported by the same geometry.

DIAGNOSE



Double edges reveal a deliberately wrong alignment.

Explain the evidence, not only the algorithm name.

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Our revised two-hour route

0–8	Problem, visual goals and one pipeline
8–35	Patch motion → gradients → Harris → feature quality
35–40	**ACTIVITY 1 · Which point can we find again?**
40–70	Patch vector → matching → 1D scale search → SIFT / ORB
70–75	Break
75–85	Distances, ratio test and ambiguous matches
85–102	Homography, RANSAC and verification
102–107	**ACTIVITY 2 · Can the fitted model mislead us?**
107–115	Warping, blending and panorama failures
115–120	Saved demonstration, lab bridge and recap

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Two five-minute activities and a five-minute break remain inside the 120-minute lesson.

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A match links the same physical point



OpenCV feature-introduction example: some small patches have many possible homes.

Coordinates can change

A scene point might move from (80, 40) to (135, 65).

Identity should not change

We want the same physical location, not merely a similar patch.

Why pixels alone struggle

Rotation, scale, lighting and occlusion alter appearance.

Similar-looking pixels are evidence, not proof of identity.

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The same building through all six steps

1 · Detect



Mark candidate locations in A and B.

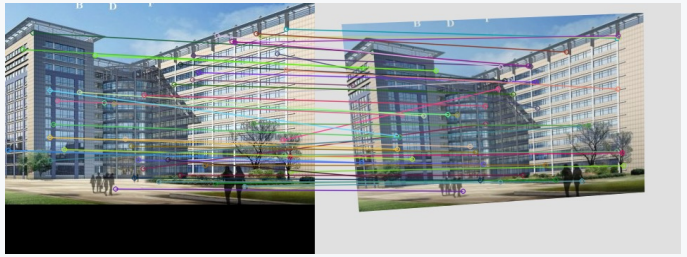
2 · Describe



patch
→ 128 values

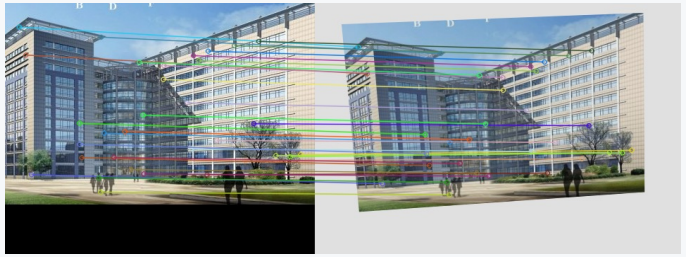
Describe a patch around each mark.

3 · Match



Compare vectors; propose pairs.

4 · Verify



RANSAC keeps a consistent subset.

5 · Warp



Map B into A's coordinate frame.

6 · Blend



Combine aligned intensities in overlap.

Controlled example: B is a transformed copy of A, not a second independently captured photograph.

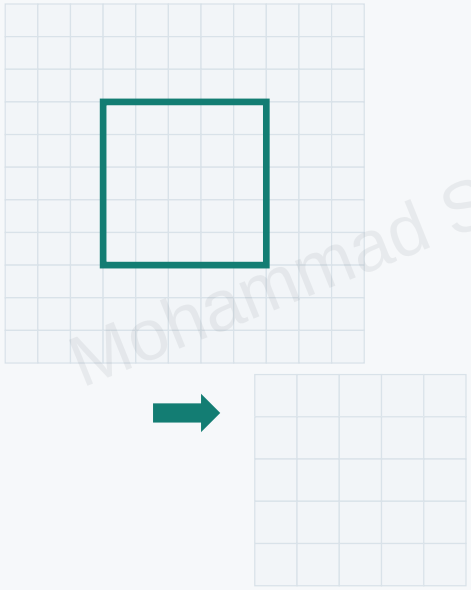
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The image stays still. We move a window.

Flat region

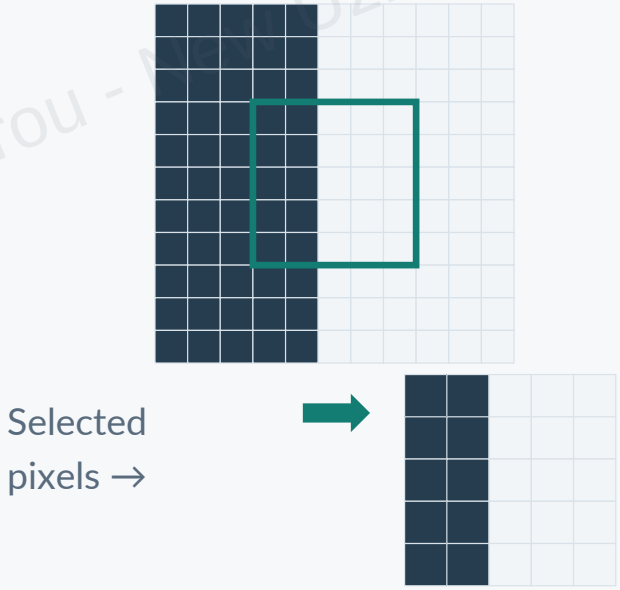
WHOLE IMAGE · simplified intensity scene



Little structure to locate.

Edge

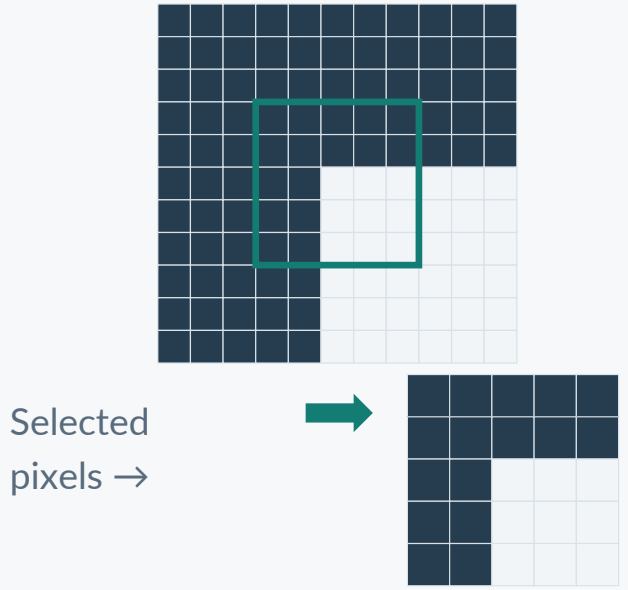
WHOLE IMAGE · simplified intensity scene



Boundary, but ambiguity along it.

Corner

WHOLE IMAGE · simplified intensity scene



Two-direction positional evidence.

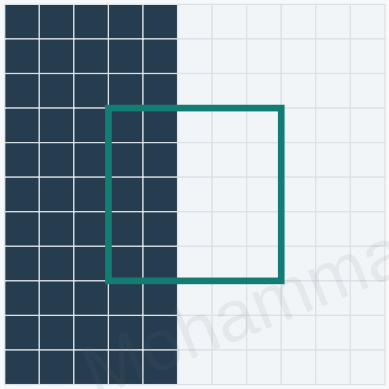
Image = the whole grid. Patch = the small group of pixels inside the outlined window.

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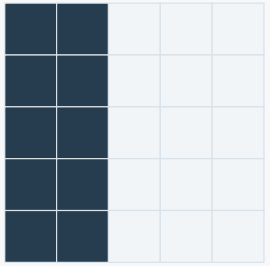
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A straight edge: right changes, down does not

Reference window

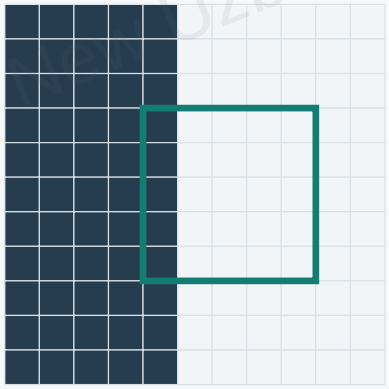


Pixels inside this window →

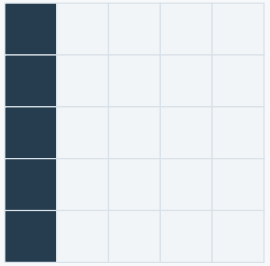


Reference

Move 1 pixel right

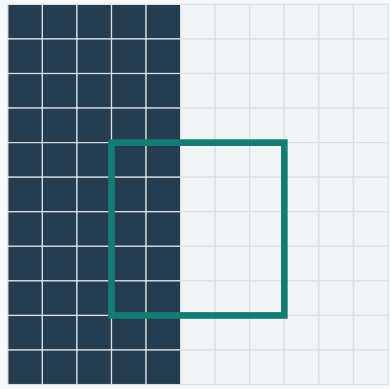


Pixels inside this window →

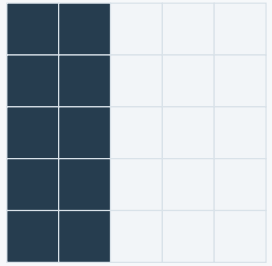


5 of 25 pixel values change

Move 1 pixel down



Pixels inside this window →



0 of 25 pixel values change

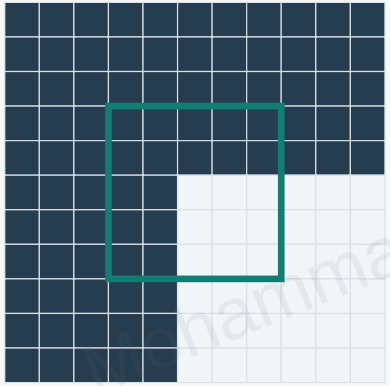
Today’s demonstration uses a 5 × 5-pixel window. This is an example size, not a Harris rule.

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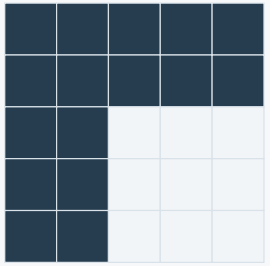
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A corner: right and down both change the patch

Reference window

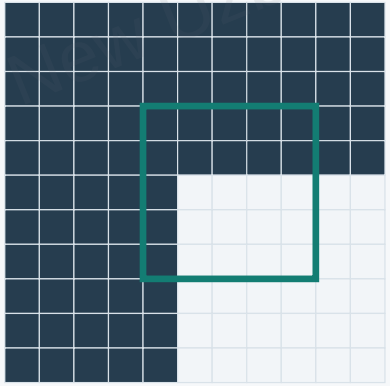


Pixels inside this window →

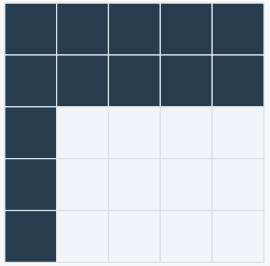


Reference

Move 1 pixel right

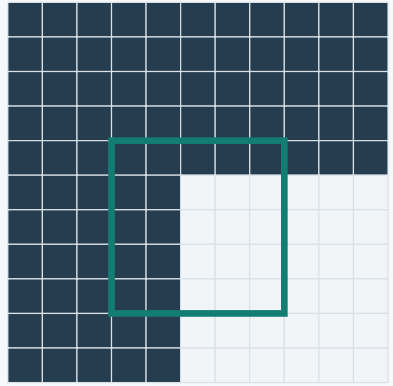


Pixels inside this window →

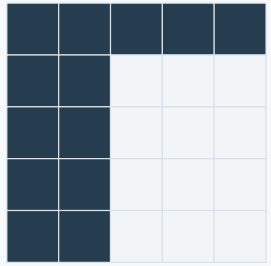


3 of 25 pixel values change

Move 1 pixel down



Pixels inside this window →



3 of 25 pixel values change

Today’s demonstration uses a 5 × 5-pixel window. This is an example size, not a Harris rule.

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Harris asks about change in two directions

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Flat

Direction 1 — **weak**

Direction 2 — **weak**

Edge

Direction 1 — **strong**

Direction 2 — **weak**

Corner

Direction 1 — **strong**

Direction 2 — **strong**

The two strengths are the eigenvalues of M; no eigenvalue calculation is needed today.

Large change in only one direction suggests an edge, not a well-localised point.

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First turn neighbouring brightness into gradients

One shared numerical recipe

I_x = (right − left) / 2
I_y = (below − above) / 2
Example: (2 − 0) / 2 = 1

0	0	0	0	0
0	0	0	0	0
0	0	2	2	2
0	0	2	2	2
0	0	2	2	2

0 = dark
2 = bright
Teal = nine evaluation sites

The outer 5 × 5 supplies neighbours.
Compute gradients at the inner 3 × 3 sites.
Use all nine sites, with weight 1.

$A = \Sigma I_x^2$ $B = \Sigma I_x I_y$ $C = \Sigma I_y^2$

I_x and I_y are brightness changes, not coordinates. We sum products of these changes.

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Flat: intensity information, little position evidence

Image patch I

2	2	2	2	2
2	2	2	2	2
2	2	2	2	2
2	2	2	2	2
2	2	2	2	2

Horizontal I_x

0	0	0
0	0	0
0	0	0

Vertical I_y

0	0	0
0	0	0
0	0	0

$$A = \Sigma I_x^2 = 0 \times 1^2 = 0$$
$$C = \Sigma I_y^2 = 0 \times 1^2 = 0$$
$$B = \Sigma I_x I_y = 0 \times 1 = 0$$

$$M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Centred differences at the nine inner sites; all weights = 1.

$\det(M) = 0 \times 0 - 0^2 = 0$ $\quad \text{trace}(M) = 0$

With $k = 0.04$: $R = 0 - 0.04 \times 0^2 = 0$

A flat region describes brightness or material, but is a poor point to localise precisely.

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Edge: strong across the boundary, weak along it

Image patch I

0	0	2	2	2
0	0	2	2	2
0	0	2	2	2
0	0	2	2	2
0	0	2	2	2

Horizontal I_x

1	1	0
1	1	0
1	1	0

Vertical I_y

0	0	0
0	0	0
0	0	0

$$A = \Sigma I_x^2 = 6 \times 1^2 = 6$$
$$C = \Sigma I_y^2 = 0 \times 1^2 = 0$$
$$B = \Sigma I_x I_y = 0 \times 1 = 0$$

$$M = \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix}$$

Centred differences at the nine inner sites; all weights = 1.

$\det(M) = 6 \times 0 - 0^2 = 0$ $\quad$ $\text{trace}(M) = 6$

With $k = 0.04$: $R = 0 - 0.04 \times 6^2 = -1.44$

An edge is useful boundary evidence; sliding along it can leave location ambiguous.

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Corner: two directions contribute to the matrix

Image patch I

0	0	0	0	0
0	0	0	0	0
0	0	2	2	2
0	0	2	2	2
0	0	2	2	2

Horizontal I_x

0	0	0
1	1	0
1	1	0

Vertical I_y

0	1	1
0	1	1
0	0	0

$$A = \Sigma I_x^2 = 4 \times 1^2 = 4$$
$$C = \Sigma I_y^2 = 4 \times 1^2 = 4$$
$$B = \Sigma I_x I_y = 1 \times 1 = 1$$

$$M = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

Centred differences at the nine inner sites; all weights = 1.

$\det(M) = 4 \times 4 - 1^2 = 15$ $\quad \text{trace}(M) = 8$

With k = 0.04: $R = 15 - 0.04 \times 8^2 = 12.44$

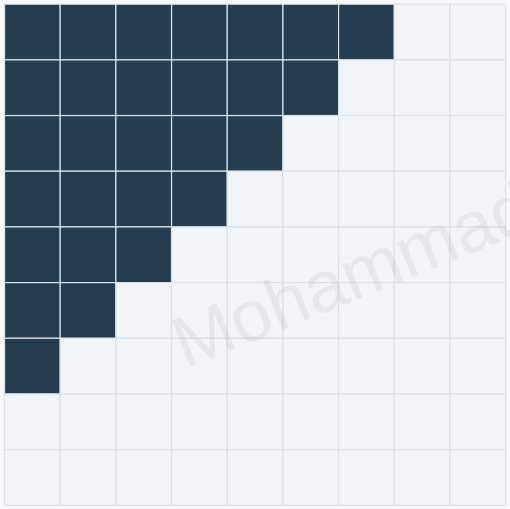
A corner supplies precise local position evidence; matching still needs a distinctive patch.

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Why do we keep the cross-term B?

A diagonal edge can fool a two-sum test



Gradients can all point in the same diagonal direction.

Example: (1, 1), repeated 3 times.

Illustrative gradient set; not a new Sobel calculation.

$A = 3 \quad C = 3 \quad B = 3$

$M = [3\ \ 3;\ \ 3\ \ 3]$

$\det(M) = 3 \times 3 - 3^2 = 0$

$\text{trace}(M) = 6$

$R = 0 - 0.04 \times 6^2 = -1.44$

The tangent direction is still weak.

Large I_x^2 and I_y^2 do not by themselves prove two independent directions.

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Why this score, rather than total change alone?

Class question: why $R = \det(M) - k \cdot \text{trace}(M)^2$?

det(M)

$\lambda_1\lambda_2$: both strengths must contribute.

trace(M)

$\lambda_1 + \lambda_2$: total directional variation.

k

Penalty strength, not confidence or a probability.

Case	det(M)	trace(M)	R at k = 0.04
Flat	0	0	**0**
Edge	0	6	**−1.44**
Corner	15	8	**12.44**

The product rewards two strong directions; the trace-squared penalty discourages imbalance.

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Harris in one compact, labelled recipe

$$R = \det(M) - k \cdot \text{trace}(M)^2$$

$$M = \begin{bmatrix} A & B \\ B & C \end{bmatrix}$$

$$A = \Sigma\, w\, I_x^2 \qquad B = \Sigma\, w\, I_x I_y$$

$$C = \Sigma\, w\, I_y^2$$

$\det(M) = A \cdot C - B^2$

$\text{trace}(M) = A + C$

$k = 0.04$ in our examples

$w = 1$ in our worked sums

Flat → weak score. Edge-like → often negative. Corner-like → sufficiently strong positive peak.

M is a summary of many nearby gradients. R is one score for the window location.

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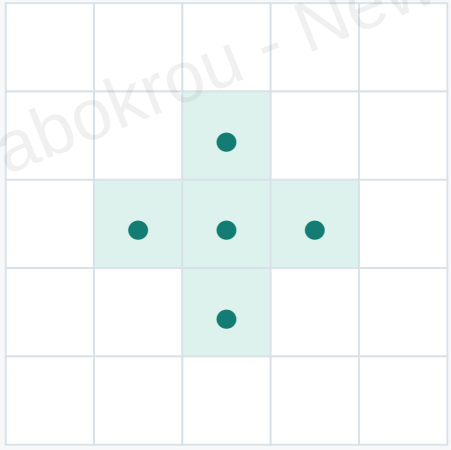
A response map is not yet a list of corners

1 · Response map

0	0	1	0	0
0	3	8	5	0
1	7	12	8	1
0	3	7	4	0
0	0	0	0	0

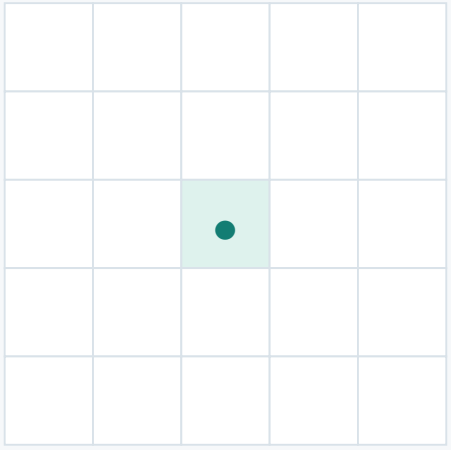
Several nearby pixels score highly.

2 · Threshold > 6



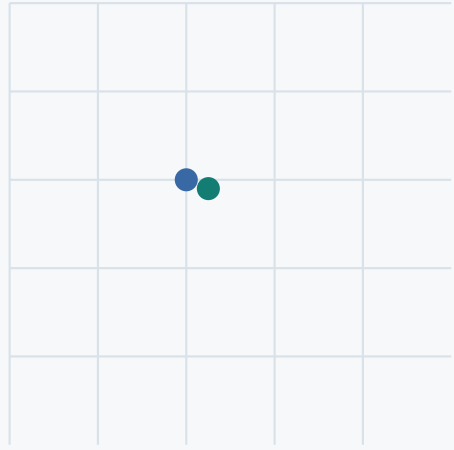
Keep strong candidates.

3 · Local maximum



Keep 12, not the entire cluster.

4 · Refine position



(2, 2) → (2.25, 2.10)

Fit a location between sample pixels.

Illustrative map and refinement coordinates; not a second measurement of the preceding toy patch.

Threshold removes weak responses; non-maximum suppression removes nearby duplicates.

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More detections does not mean better features



Would 500 points on the same repeated windows be enough?

Repeatability

Does the same scene structure reappear across views?

Distinctiveness

Does its surrounding patch differ from competing patches?

Localisation accuracy

Are the detected coordinates close to the true image position?

Spatial distribution

Do useful points cover the overlap, rather than one tiny cluster?

Judge repeatability, distinctiveness, localisation accuracy and spatial distribution—not count alone.

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A point locates. A surrounding patch describes.

The descriptor comes from the neighbourhood around the keypoint—not the corner pixel alone.





[18, 0, 5, …]
appearance vector

Dot = one position.

Outline = nearby pixels used for description.

Detector	Descriptor	Matching	Verification
Find a location.	Summarise a patch.	Compare vectors.	Check geometry.

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Which point can we find again?

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REASONING QUESTION · 90 SECONDS

Two photographs show the same wall after a small camera movement. Which patch is easiest to relocate precisely?

A · BLANK WALL **B · STRAIGHT VERTICAL EDGE** **C · ISOLATED L CORNER**

Give one reason that compares a small horizontal shift with a small vertical shift. Answer in two sentences, using your own reasoning.

Class Engage · CE 8480 · Use your registered email

Initial answer: 90 seconds. Choose a patch and explain your reasoning in two sentences.

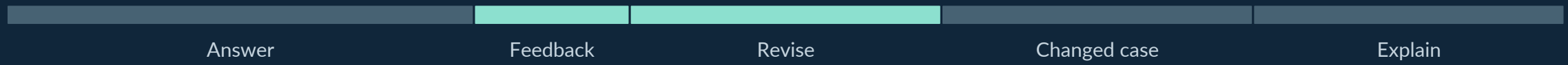
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Read feedback. Improve the reason.

FEEDBACK 30S → REVISION 60S

Compare moving a patch along an edge with moving it across the edge. Which movement could leave the patch looking almost unchanged? Check that your explanation accounts for both horizontal and vertical position, rather than only brightness or the strength of one intensity change.

Read the feedback, then revise or keep your choice. State what changed in your explanation, or why your original reasoning still holds. One or two sentences are enough.



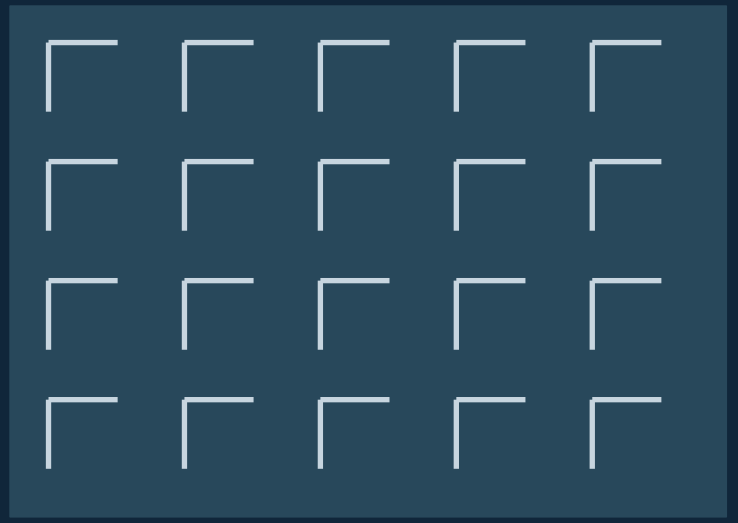
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Changed case: twenty identical corners

INDEPENDENT CHANGED CASE · 60 SECONDS

Changed case — work independently. The L-shaped corner is now one of twenty identical corners in a tiled wall. A detector finds every corner very accurately. Does this guarantee that the same physical corner can be identified in the other photograph? Explain, and name one extra visual cue that would help.



20 repeated corners · schematic

Answer | Feedback | Revise | Changed case | Explain

Class Engage · CE 8480 · Use your registered email

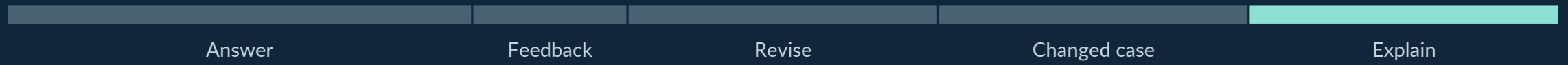
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Localisation is not identity

EXPLANATION · FINAL 60 SECONDS

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Initially, C is easiest to localise: moving the small window horizontally or vertically changes the corner pattern. A vertical edge leaves movement along the edge uncertain; a flat patch leaves both directions uncertain. In the changed case, all twenty corners can be well localised but still look alike. Localisation and identity are different problems. A distinctive surrounding mark or wider context helps identify the same physical location.



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We found a corner. What do we do with it?

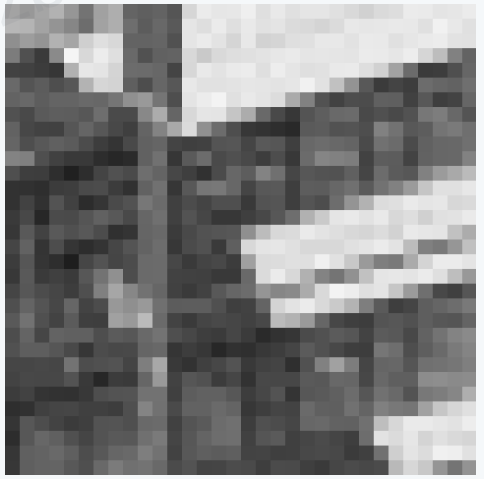
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1 DETECT A LOCATION



Known reference photograph

2 TAKE THE SURROUNDING PATCH



Centre: (230, 91)

Window: 32 × 32 pixels

Use all nearby pixels—not just the intensity of the corner pixel.

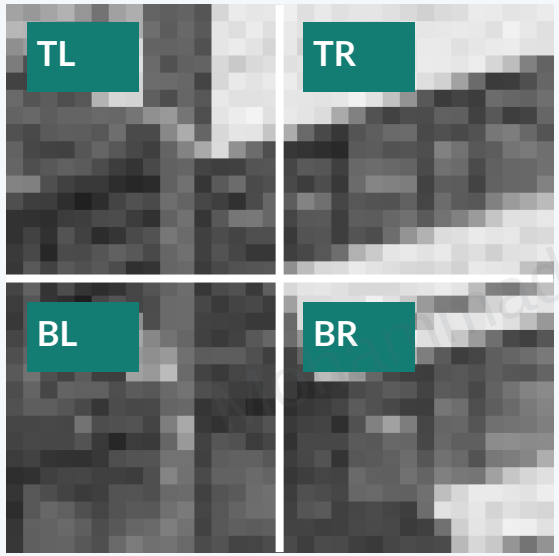
Next: turn this patch into a short list of measurements.

The dot gives a location. The neighbouring pixels provide the appearance evidence.

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A simple descriptor: four average brightness values

SAME 32 × 32 PATCH



Each region is 16 × 16:
256 grayscale pixel values.

FEATURE = MEAN BRIGHTNESS IN ONE REGION

Top-left	29,522 ÷ 256	**≈ 115**
Top-right	39,957 ÷ 256	**≈ 156**
Bottom-left	21,381 ÷ 256	**≈ 84**
Bottom-right	33,499 ÷ 256	**≈ 131**

d = [115, 156, 84, 131]

Fixed order: top-left, top-right, bottom-left, bottom-right. Rounded means; grayscale range 0–255.

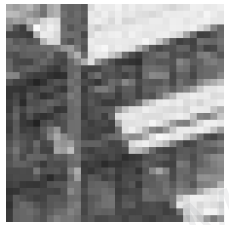
4 regions × 1 average = 4 features. This teaching descriptor is NOT SIFT.

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Compare vectors: which patch looks most similar?

Reference vector d = [115, 156, 84, 131]

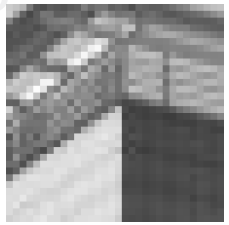
B1 · same point



[119, 160, 88, 135]

Distance = 8.0

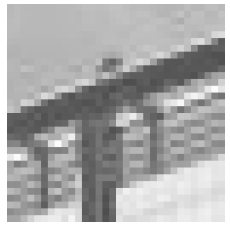
B2 · another corner



[140, 141, 187, 83]

Distance = 117.3

B3 · another corner



[168, 157, 137, 188]

Distance = 94.2

For B1: $D^2 = (115-119)^2 + (156-160)^2 + (84-88)^2 + (131-135)^2$

Each difference is −4, so $D^2 = 16 + 16 + 16 + 16 = 64$. $D = \sqrt{64} = 8$.

Nearest descriptor = a candidate match. It is not yet proof of the same physical point.

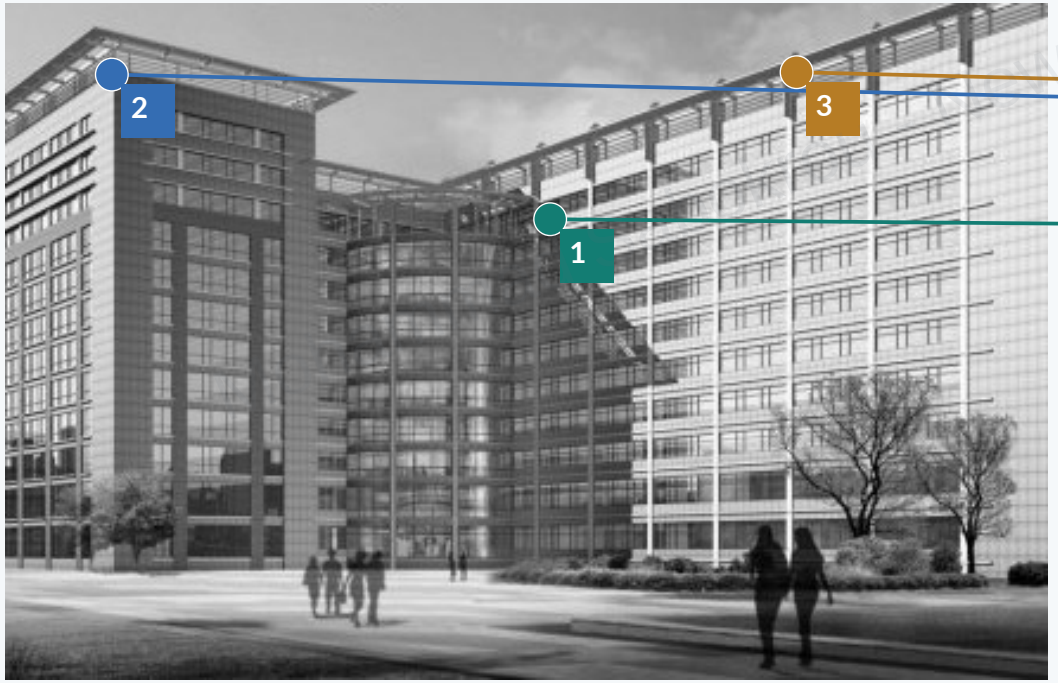
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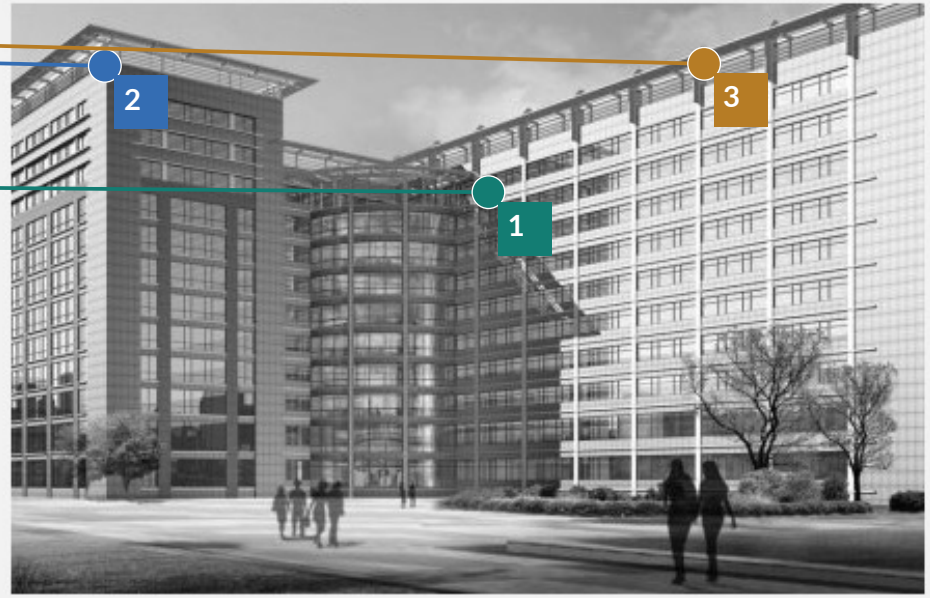
Several matched patches support the same image identity

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KNOWN REFERENCE: THIS BUILDING



QUERY: TRANSLATED + 4 GRAY LEVELS



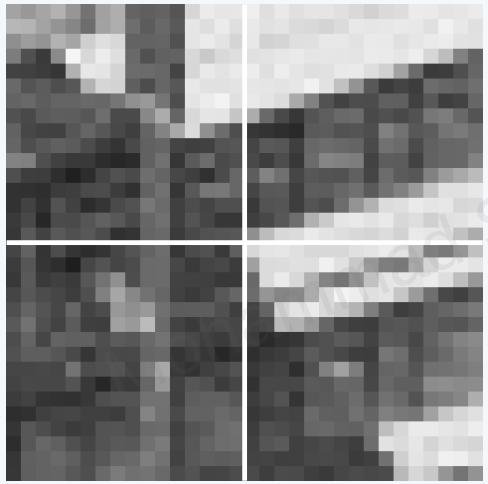
All three pairs agree with the same shift: (+28, +18) pixels. Later, we test many pairs and reject bad ones.

The reference supplies the name. Local matches supply evidence—not semantic understanding.

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Same place—but our four-number vector changes

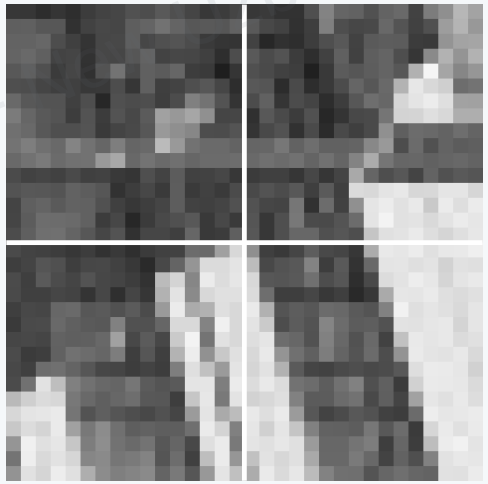
Reference



[115, 156, 84, 131]

D = 0

Rotate 90° clockwise

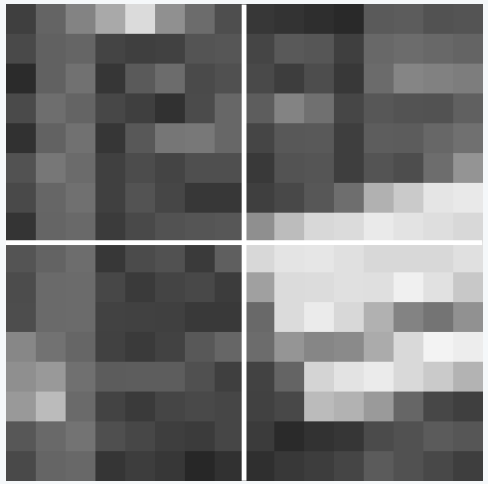


[84, 115, 131, 156]

D = 74.0

Rotation swaps where texture falls.

Zoom ×2; same window



[88, 110, 84, 150]

D = 56.6

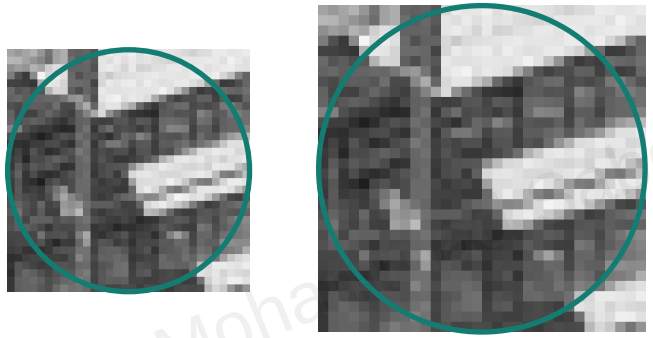
A fixed window covers less scene.

Even with the correct centre supplied, this fixed, axis-aligned descriptor is not invariant.

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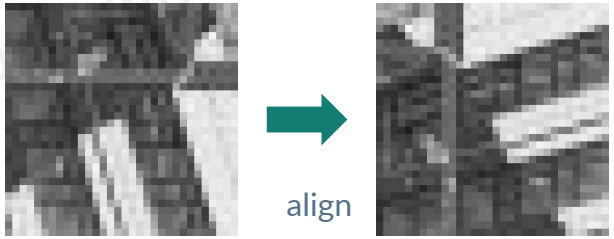
SIFT improves the description—not the idea of matching

1 · Choose a useful scale



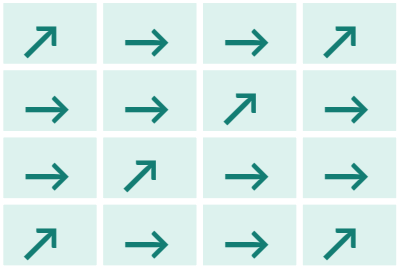
Describe a comparable amount of the scene.

2 · Align the local direction



Describe relative to the feature, not the image axes.

3 · Summarise gradients



4 × 4 cells × 8 directions = 128 values

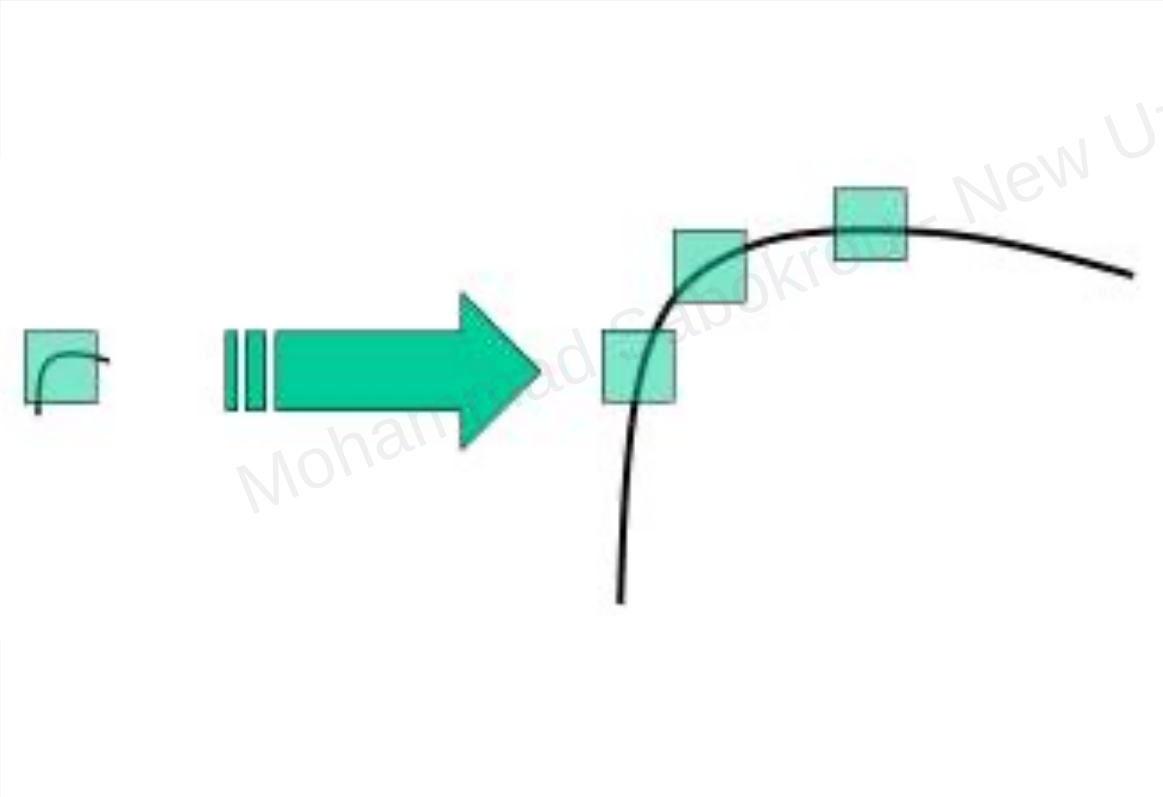
Use local edge directions, not raw brightness alone.

Detect with SIFT → describe the neighbourhood → compare vectors → verify geometry

We still compare vectors. SIFT makes local descriptions more comparable across scale and rotation.

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A fixed-size window does not fit every scale



OpenCV SIFT introduction: the same fixed window sees different structure after enlargement.

Move closer

A small feature occupies more pixels.

Keep the same window

It may cover only a smooth part of a larger structure.

Better approach

Search over multiple sizes and record a useful scale.

Scale should be part of the feature, not an accidental camera setting.

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A keypoint has a small “passport”



position · radius · direction

x, y

The position in image pixels.

Scale

How large a neighbourhood to describe.

Orientation

Which direction counts as the local reference.

Descriptor

A separate numerical summary of that neighbourhood.

Position locates the patch; scale and orientation define its local frame.

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SIFT in four plain-language steps

1 · Search sizes

Find candidates over image position and scale.

2 · Keep stable points

Refine location; reject weak and edge-like candidates.

3 · Pick a direction

Assign a dominant local orientation.

4 · Describe the patch

Record the distribution of local gradient directions.

SIFT detects features and describes them; it is not just “Harris plus a vector.”

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Find a useful point—and its scale

Imagine one grayscale row through a small bright structure.

Position x	0	1	2	3	4	5	6	7	8
Intensity	0	0	0	100	100	100	0	0	0

Position: WHERE?

Which sample gives a useful centre?
The three bright samples are all 100.

Scale: HOW BROAD?

Try σ = 0.5, 1, 2, 4 pixels.
Larger σ means a wider Gaussian blur.

We will search position AND blur scale. This 1D example has no literal corner.

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What does one Gaussian blur calculate?

Compute the centre x = 4 at Gaussian scale σ = 1.

Input position	x = 3	x = 4	x = 5
Input intensity	100	100	100
Gaussian weight	0.24197	0.39894	0.24197

$$L(4, 1) = 100 \times 0.24197 + 100 \times 0.39894 + 100 \times 0.24197$$

L(4, 1) ≈ 88.29

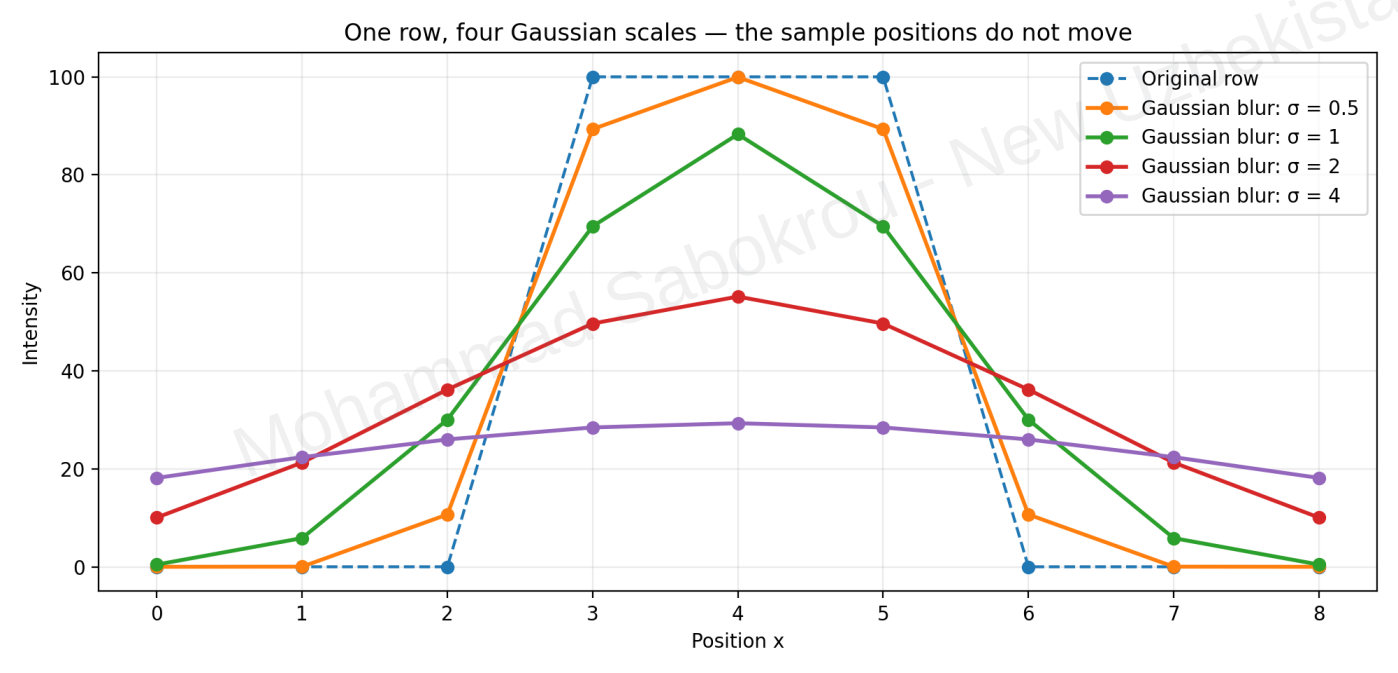
L(x, σ) means the blurred intensity at position x.
All other input samples are 0, so their contributions are 0.
Weights are rounded here; the full normalized kernel is used.

A Gaussian blur is a weighted average. Nearby samples receive larger weights.

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Repeat the blur at four scales



Compute each blur from the same original row.
Values outside the displayed row are zero.

Three central positions

σ	x=3	x=4	x=5
0.5	89.33	**99.95**	89.33
1	69.49	**88.29**	69.49
2	49.65	**55.15**	49.65
4	28.44	**29.31**	28.44

The centre falls from 99.95 to 29.31 as the bright region is spread over more positions.

Same row. Same positions. Different blur widths—not four different photographs.

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Subtract neighbouring Gaussian scales

$$D(x, \sigma) = L(x, 2\sigma) - L(x, \sigma) \quad \text{[coarse teaching scale ratio = 2]}$$

At centre x = 4	More blurred	Less blurred	Difference
σ: 0.5 → 1	88.288	99.947	-11.66
σ: 1 → 2	55.154	88.288	**-33.13**
σ: 2 → 4	29.307	55.154	-25.85

Example: 55.154 − 88.288 ≈ −33.13. Keep the negative sign.

Four blurred rows produce three difference rows; calculations use full precision.

Difference of Gaussians (DoG) measures change between blur levels—not raw brightness.

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Does −33.13 stand out in position AND scale?

Compare the centre with all eight surrounding entries.

DoG scale pair	x = 3	x = 4	x = 5
0.5 → 1	-19.84	-11.66	-19.84
1 → 2	-19.84	**-33.13**	-19.84
2 → 4	-21.21	-25.85	-21.21

A local minimum

−33.13 is smaller than every neighbour.

2 beside it + 3 above + 3 below = 8.

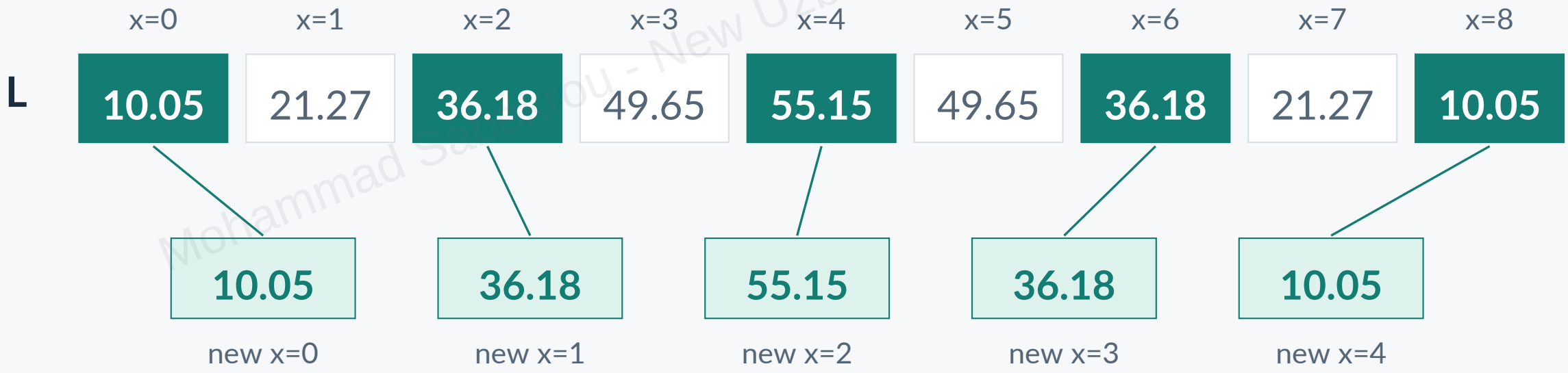
Candidate: x = 4, DoG interval σ = 1 → 2. Refine and test stability next.

Keep a local maximum OR minimum. This gives a candidate location and scale—not a descriptor.

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Blur levels and octaves are different steps

Example octave hand-off: take L(x, 2), then keep every second sample.

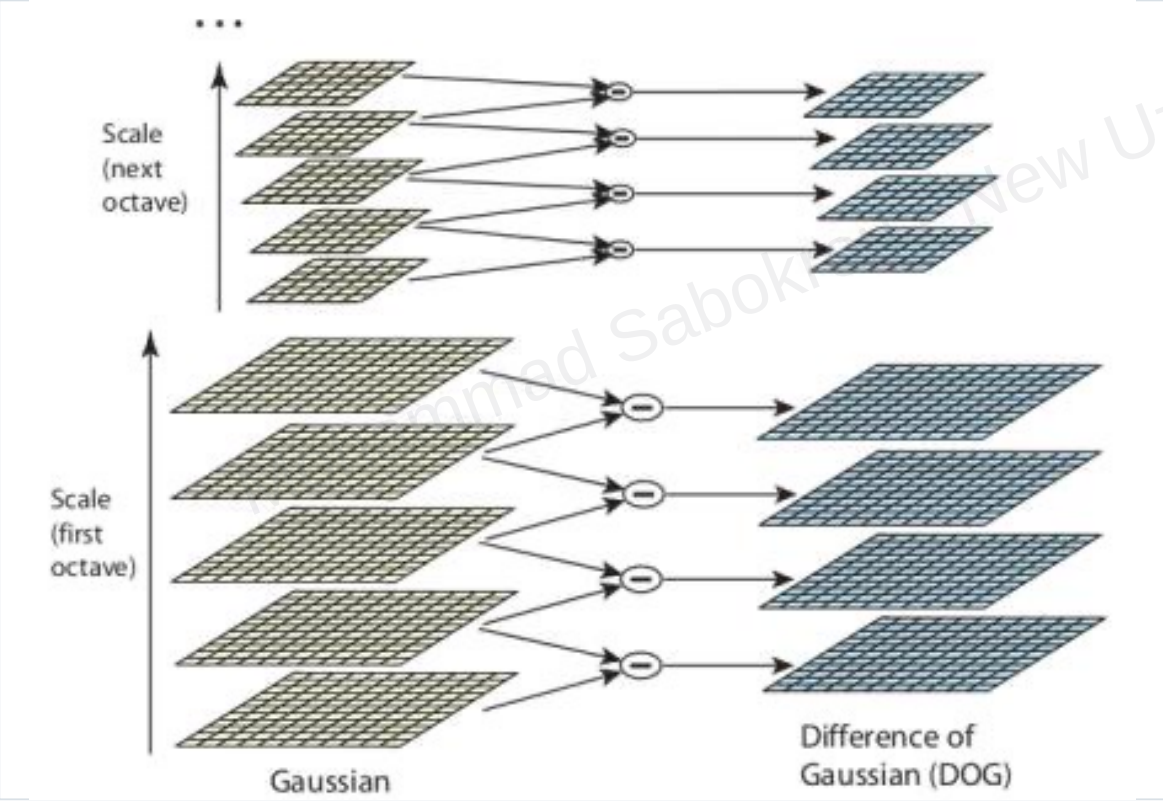


One new pixel spans two old pixels: σ = 2 old pixels becomes σ = 1 new pixel.
Real SIFT uses closer blur levels, e.g. 1.60 → 2.02 → 2.54 → 3.20 (three steps per octave).

Inside an octave: change blur. At an octave boundary: change the sampling resolution.

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Look at a family of blurred images



Gaussian scale space and differences of nearby blur levels. OpenCV illustration.

Gaussian images

Smooth the image by different amounts.

Difference of Gaussians

Subtract neighbouring blur levels.

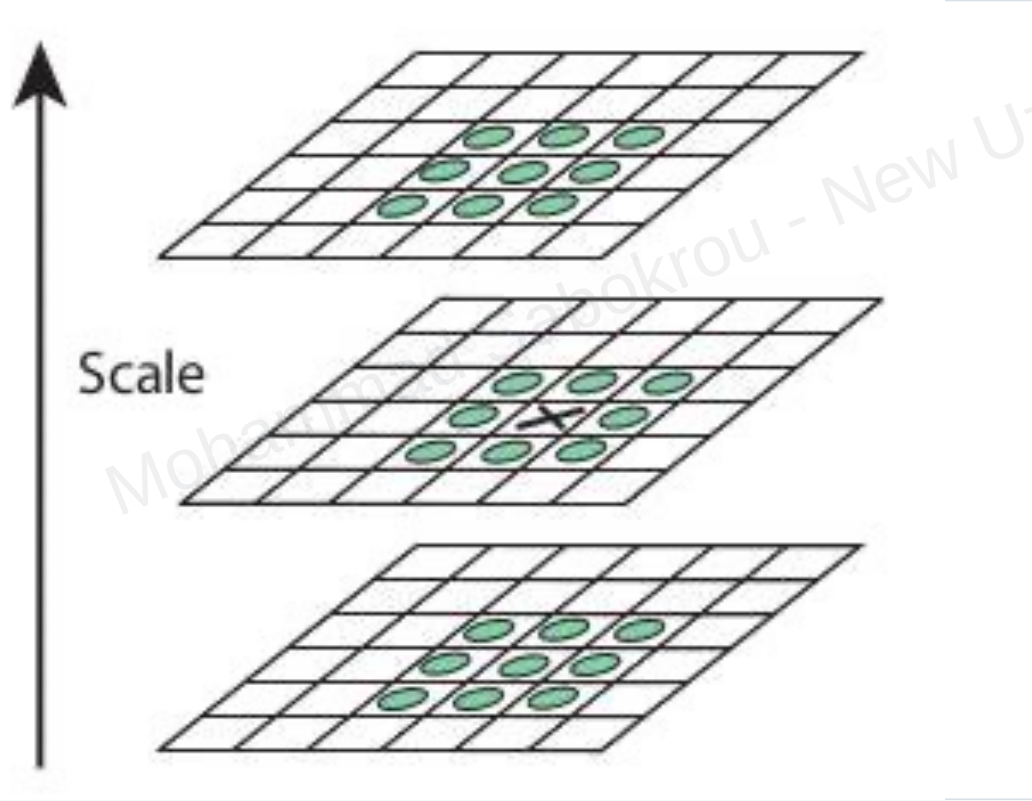
Octaves

Repeat at a smaller image size to cover larger structures.

The useful question is “at what size is this structure distinctive?”

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A candidate must stand out across space and scale



A candidate is compared with neighbours in its own level and adjacent levels.

Across space

Compare nearby pixels in the current level.

Across scale

Also compare the levels above and below.

Then check stability

Reject weak contrast and poorly localised edge responses.

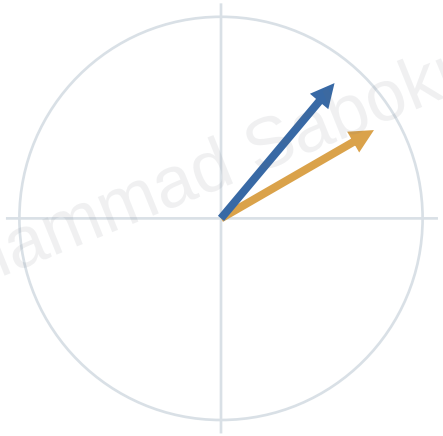
A local extremum is a candidate, not automatically a final keypoint.

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Describe directions relative to the feature

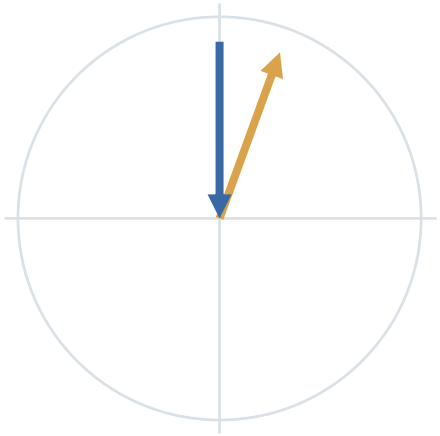
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Original patch



50° − 30° = 20°

Image rotated by 40°

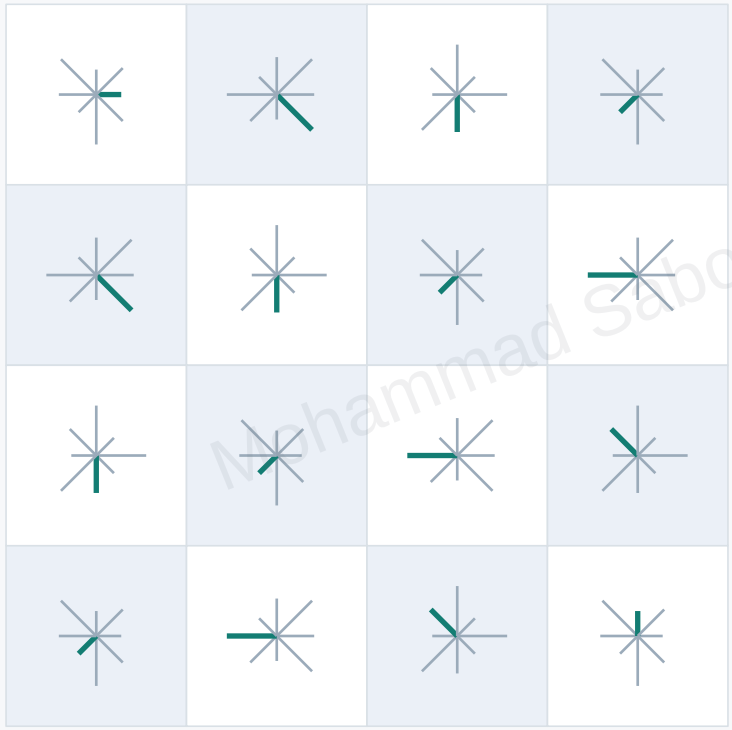


90° − 70° = 20°

Subtracting the local reference direction can cancel a shared rotation.

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A SIFT descriptor is a pattern of directions



16 spatial cells

8 bins per cell

128

descriptor values

Schematic gradient bins; not measured data.

4 × 4 spatial cells × 8 orientation bins = 128 values.

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Normalisation reduces some lighting effects

Before gain

$$(3, 4) \div 5 = (0.6, 0.8)$$

After ×2 gain

$$(6, 8) \div 10 = (0.6, 0.8)$$

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More tolerant does not mean immune to shadows, clipping or occlusion.

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ORB asks many tiny yes/no questions

ONE COMPARISON



p = 30 q = 90

30 < 90 → 1

Find and orient the patch

FAST finds candidate points; a pyramid handles size. ORB assigns orientation.

Compare two sample locations

In the oriented patch, is intensity at p smaller than at q?

Store binary answers

30 < 90 → 1. Repeating selected tests makes the descriptor.

ORB = Oriented FAST and Rotated BRIEF. Default: 256 bits (32 bytes), Hamming distance.

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Choose a descriptor and the matching distance together

Method	Description	Distance	Practical focus
SIFT	128 numeric values	L2	Scale/orientation-normalised gradients
ORB (default)	256 bits / 32 bytes	Hamming	Compact binary tests; efficient matching

Use the descriptor’s meaning, not merely the array data type, to choose a metric.

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Five-minute break

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05:00

Next: how do we decide whether two descriptions belong together?

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Distance means “how different?”

L2 · numerical vectors

(1, 2) and (4, 6)

$\sqrt{(3^2 + 4^2)} = 5$

Hamming · binary tests

10110010 and 10010110

2 differing bits

Small descriptor distance suggests similarity, not guaranteed correspondence.

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Nearest neighbours propose candidates

Query descriptor

One patch in image A

Image B candidates

distance 20 · distance 40 · distance 75

Candidate match

Choose the nearest, then ask whether it is distinctive.

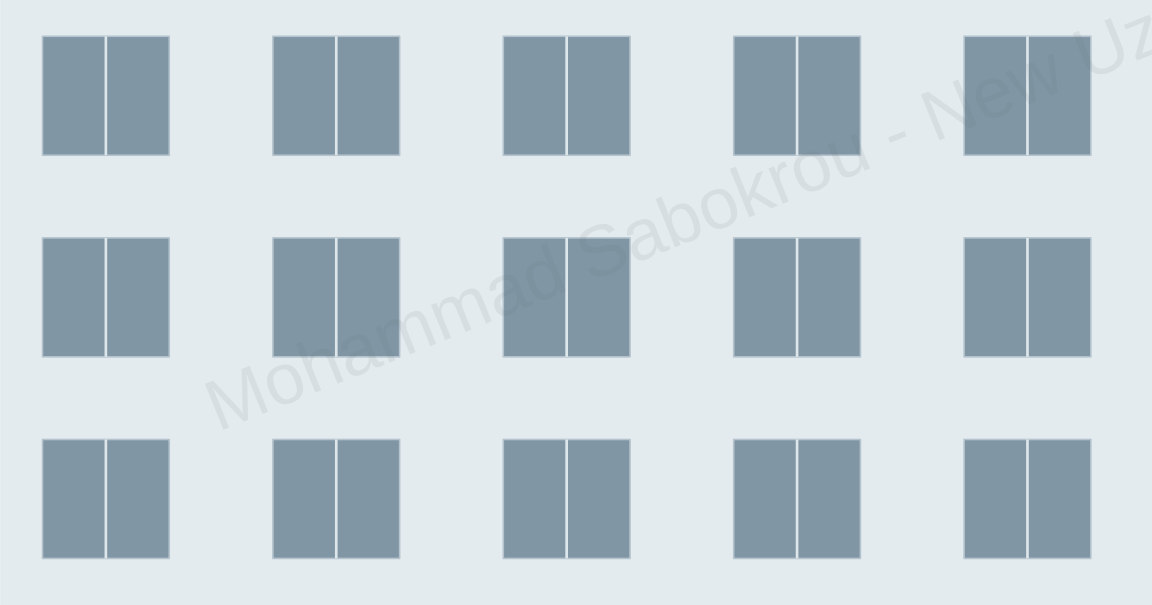


Candidate lines in an OpenCV tutorial example.

Keep the best and second-best options so that you can inspect ambiguity.

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Repeated windows create lookalikes



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Candidate 1

$d_1 = 40$

Candidate 2

$d_2 = 42$

Interpretation

Two almost equally plausible descriptions.

“Best” is weak evidence when another candidate is almost as good.

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The ratio test asks for a clear winner

d_1 = nearest distance d_2 = second-nearest distance

d_1	d_2	Ratio	Decision
20	40	0.50	Keep
40	45	0.89	Reject
15	20	0.75	Reject (strict <)
3	20	0.15	Keep

Accept only when $d_1 / d_2 < 0.75$ in this worked example.

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Cross-check asks whether the choice is mutual

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A

A local descriptor

→

B

Its preferred candidate

A chooses B

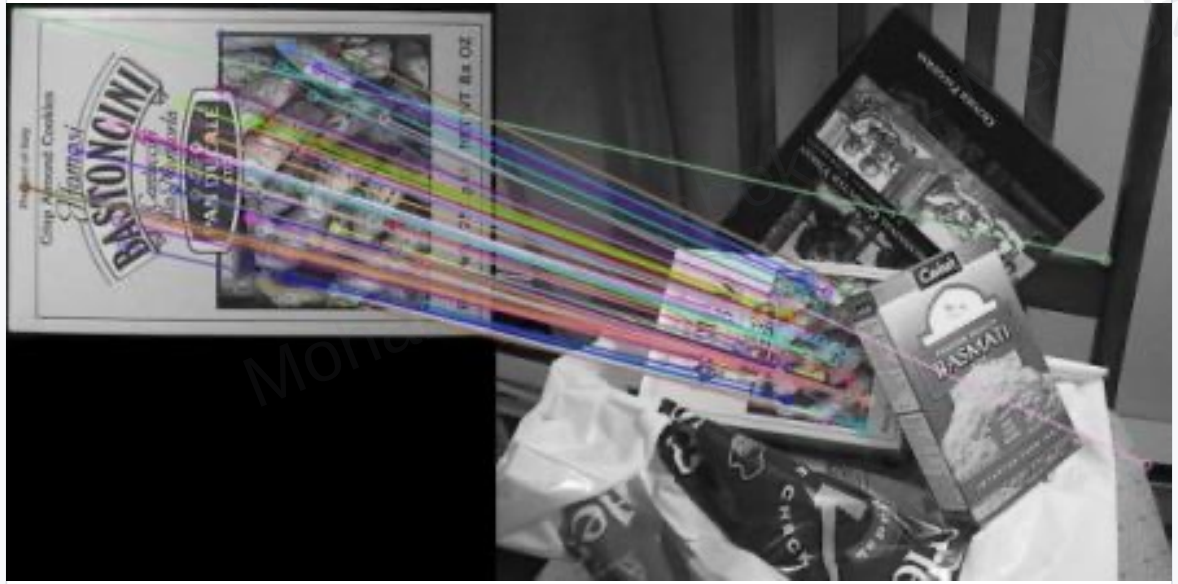
←

B chooses A

Mutual nearest neighbours can still be a mutually wrong pair.

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Filtered matches are still hypotheses



OpenCV SIFT ratio-filtered matches; lines are candidate correspondences.

A useful filter

Remove many ambiguous descriptor matches.

A remaining question

Do the point locations agree with plausible geometry?

What not to conclude

A page full of lines is not an accuracy measurement.

Descriptor evidence and geometric evidence answer different questions.

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Before blaming geometry, check the earlier stages

What you observe	Possible cause	Useful check
Very few keypoints	Low texture or heavy blur	Inspect the image and keypoint map
Many points, few matches	Wrong metric / limited overlap / large change	Check the metric and shared scene
Many similar candidates	Repeated appearance	Look for distinctive context

Count → inspect → form a hypothesis → change one thing.

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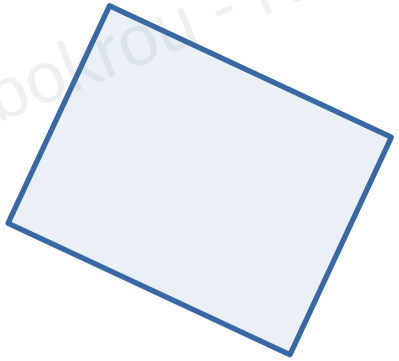
Geometry describes how points move together

Translation



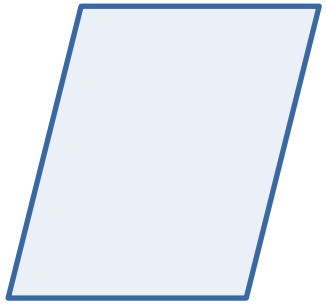
2 free parameters

Similarity



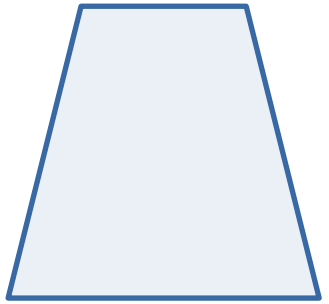
4 free parameters

Affine



6 free parameters

Homography



8 free parameters

Choose a transformation because of the scene, not because it has more parameters.

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When can one homography explain the image pair?

Good model: a plane

A poster, a page or a nearly flat wall.

Good model: pure rotation

The camera rotates about a fixed optical centre.

Warning: depth + translation

Nearby and distant objects can move differently.

A flat surface or pure camera rotation is the key ideal case.

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Homography: a three-by-three mapping recipe

$$H = \begin{bmatrix} h11 & h12 & h13 \\ h21 & h22 & h23 \\ h31 & h32 & h33 \end{bmatrix}$$

[x, y, 1] → [a, b, c]

Output point: (a/c, b/c)

hij: a coefficient in row i, column j of H
x, y: input coordinates · a, b, c: the matrix-product outputs
c must be nonzero for a finite output point.

Multiply first. Divide by the third coordinate second.

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Two small numerical examples

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TRANSLATION

$$\begin{bmatrix} 1 & 0 & 20 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(30, 40) \rightarrow (50, 50)$$

PROJECTIVE DIVISION

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0.01 & 0 & 1 \end{bmatrix}$$

$$(100, 50) \rightarrow (100, 50, 2) \rightarrow (50, 25)$$

The final division matters whenever the third output is not 1.

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Four pairs are a minimum, not a quality guarantee

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Well spread



Constrains a 2D region.

On one line



Does not determine the mapping away from the line.

For a minimal four-pair estimate, use general position: no three collinear.

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A few bad pairs can pull an ordinary fit away

Candidate matches

A mixture of useful pairs and bad pairs.

Fit everything equally

Large errors can distort the estimate.

Seek consensus

Find a plausible model supported by many pairs.

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We need a method that can fit a model despite some incorrect candidates.

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RANSAC: sample, fit, test, repeat

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1 Sample 4 suitable pairs → **2** Fit a homography → **3** Test all candidate pairs → **4** Repeat and compare support → **5** Refit using the inliers

A consensus is a hypothesis to inspect, not an unconditional truth guarantee.

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Reprojection error measures a geometric miss

Error = distance between the observed point and H’s prediction

Observed	Predicted	Error (px)	At < 3 px
(50, 50)	(50, 50)	0	Inlier
(22, 12)	(20, 10)	2.83	Inlier
(42, 20)	(30, 20)	12	Outlier
(83, 64)	(80, 60)	5	Outlier

Here, error < 3 pixels means “inlier to this model.”

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Trust several kinds of evidence together

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1 Support

How many candidate pairs fit?

2 Coverage

Do inliers span the overlap?

3 Residuals

How far are predicted points from observations?

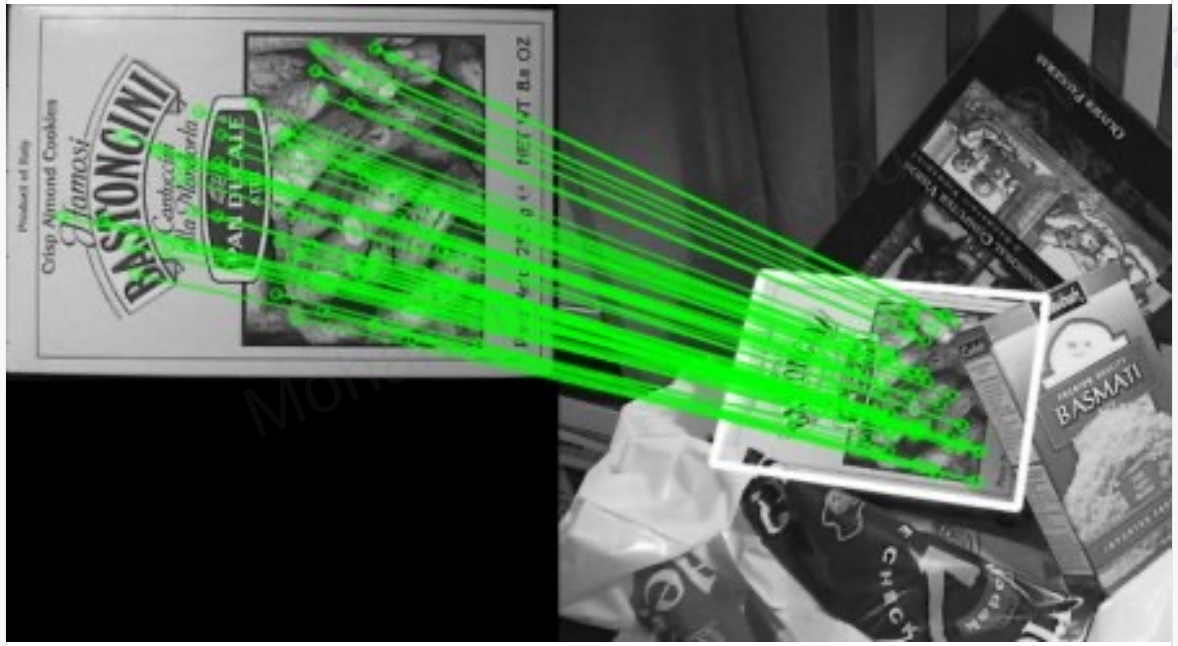
4 Plausibility

Do scene assumptions and unique landmarks agree?

High support + good spread + low error + plausible assumptions—not one number alone.

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The geometric model predicts a boundary



OpenCV homography example: the reference boundary is projected into the scene.

What H does

Maps the corners of the reference image into the scene.

What the outline means

A model prediction supported by selected matches.

What still needs checking

Correct object identity, coverage and physical plausibility.

This resolves the opening example—but does not remove the need to verify it.

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Can 80% inliers still be wrong?

INITIAL ANSWER · 90 SECONDS

A panorama uses repeated, identical windows. RANSAC finds that 60 of 75 candidate matches fit one homography within 3 pixels: an 80% inlier fraction. However, the panorama appears shifted by one window. A student says, “80% inliers proves the alignment is correct.” Do you agree? Give a reason and one independent visual check. Answer in two sentences.

80%

60 of 75 pairs fit one model

Answer | Feedback | Revise | Changed case | Explain

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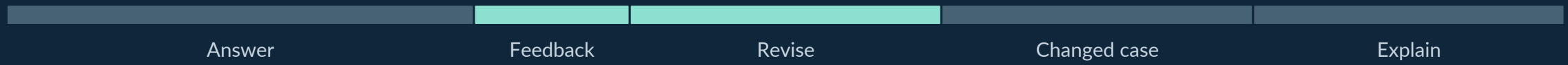
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Check what the percentage actually proves.

FEEDBACK 30S → REVISION 60S

Ask what the 80% actually measures: agreement with a model, or proof that each pair represents the same physical window? Imagine the whole repeated pattern shifted by one window. Identify a non-repeating landmark or boundary that could test the proposed alignment independently of those repeated matches.

Revise or keep your decision. State what the 80% tells you and name the independent evidence you would check. Explain any change in your reasoning.



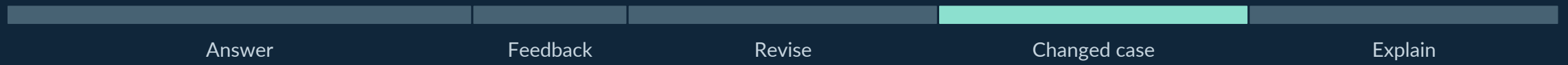
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Changed case: a nearby pole and a distant wall

INDEPENDENT CHANGED CASE · 60 SECONDS

Changed case — work independently. The matches are now physically correct. A nearby pole and a distant wall are photographed while the camera moves sideways. The wall aligns well, but the pole appears twice. Can one homography necessarily align both depths? Name one change to image capture that would reduce the problem.



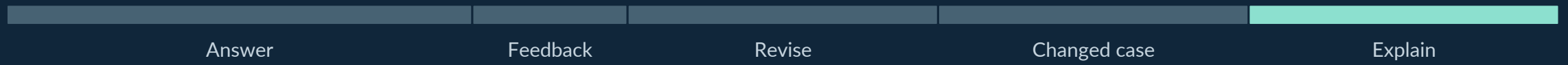
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Wrong identity and wrong geometry are different

EXPLANATION · FINAL 60 SECONDS

The 80% describes agreement with one model, not proof of correct identity. Identical windows may agree on the wrong offset; check a unique landmark and whether the alignment is physically plausible. In the changed case, correct matches still do not make one homography suitable for a translated camera viewing multiple depths. Parallax can double the pole while the wall aligns. Rotate with minimal translation or use a nearly planar/distant scene.



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From verified matches to a panorama

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Capture overlap

Both images must share useful scene content.

Estimate H

Use filtered matches and robust geometry.

Choose a canvas

Include the transformed image boundaries.

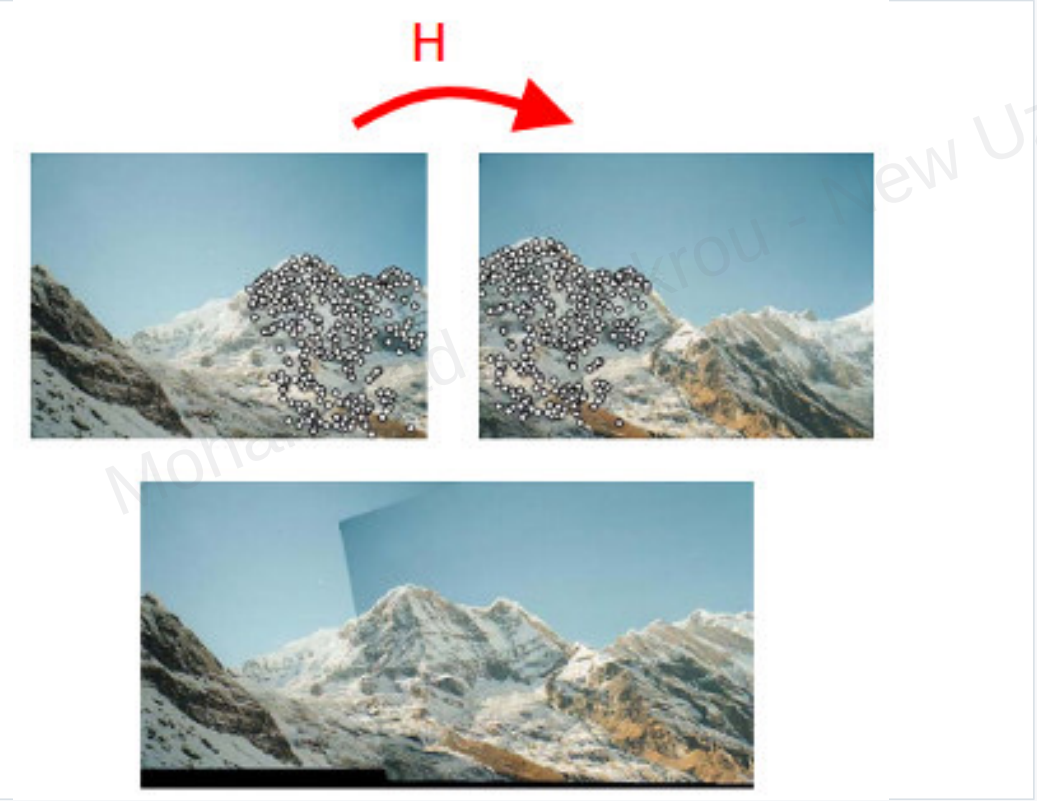
Warp and blend

Resample, combine overlap and inspect the result.

Align geometry first; combine appearance second.

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Overlapping views become a wider scene



OpenCV homography tutorial panorama example; not a student experiment.

Shared content

The overlap links the coordinate systems.

New content

Each view adds another part of the scene.

Inspection

Look at boundaries and structures in the overlap.

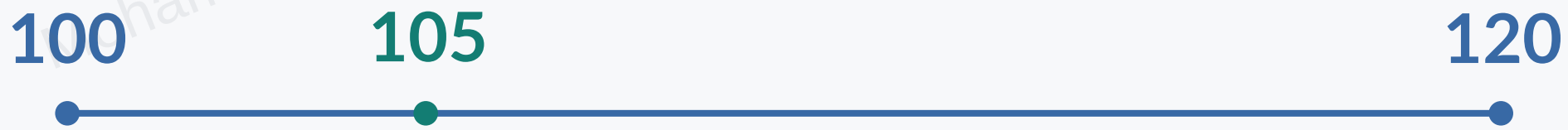
A wider image is useful only when shared structures are aligned sensibly.

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Warping also means resampling pixels

Output pixel → inverse map → source location → interpolate

One-dimensional example



¼ of the way between samples

Geometry gives a sampling location; interpolation gives an intensity.

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Blending can soften an exposure seam



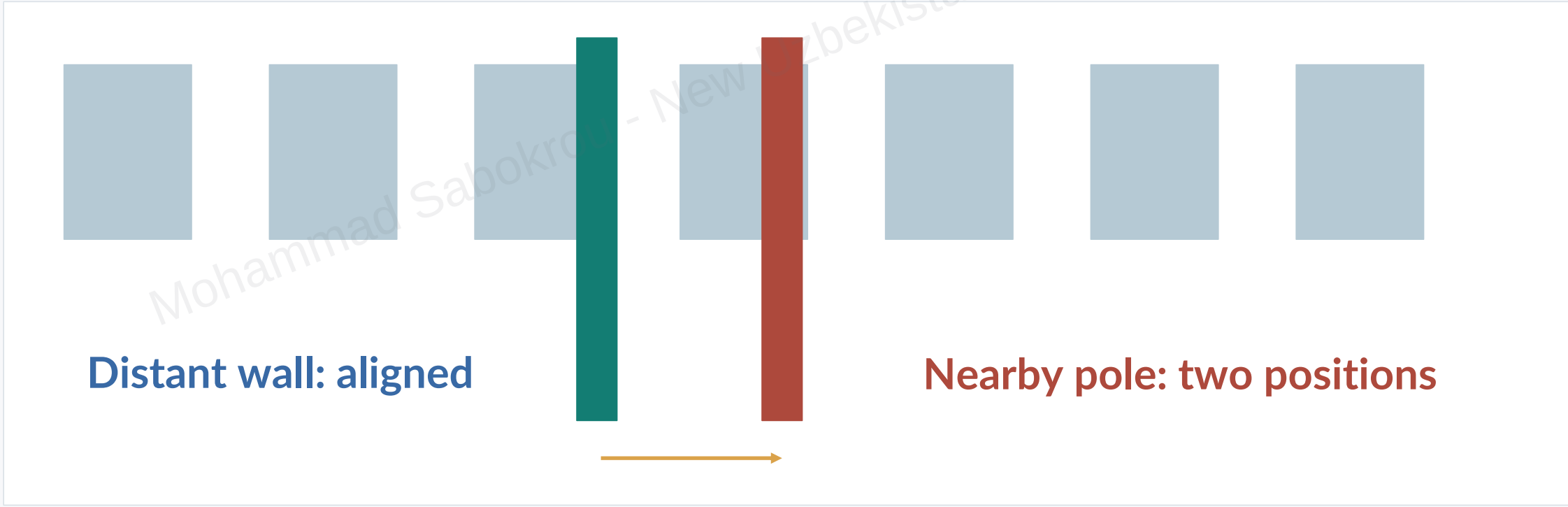
At equal weights: 0.5 × 100 + 0.5 × 140 = 120

A smooth brightness transition is not a geometric accuracy test.

Blending can soften a brightness jump. It cannot repair mismatched geometry.

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Parallax: the wall aligns, the pole doubles



Schematic overlap after fitting the wall

Depth variation plus camera translation can violate the one-homography model.

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Diagnose the failure before choosing the repair

Symptom	Likely issue to inspect	A targeted repair
Few usable points	Low texture / blur	Add textured context; hold steady
One-window offset	Repeated appearance	Include a unique landmark
Doubled near object	Parallax / wrong model	Reduce translation or depth variation
Brightness seam	Exposure mismatch	Stabilise exposure; blend after alignment
Only moving objects double	Scene motion	Capture a static scene / retake

A repair should address the cause, not merely make the output look smoother.

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Capture a pair that gives matching a fair chance

Share useful content

Keep substantial overlap and distinctive texture.

Stabilise capture

Hold steady; avoid motion blur and changing exposure.

Respect the model

Prefer a plane or near-pure rotation.

Keep it safe

Use permitted scenes; avoid unnecessary personal information.

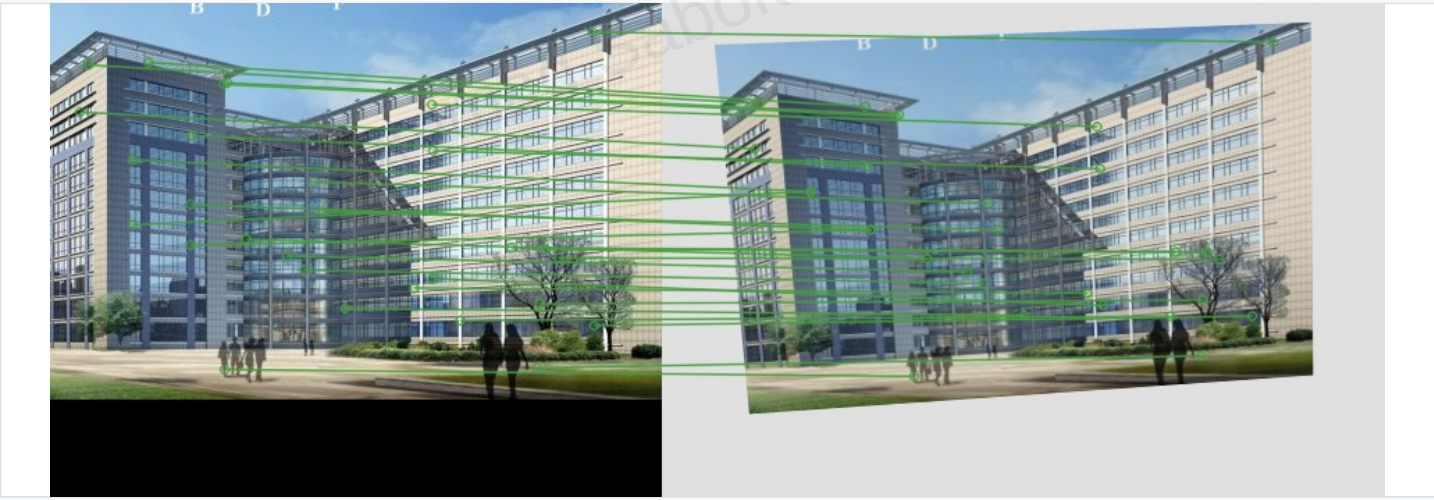
For a comparison, change one condition deliberately and keep the others similar.

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A measured walk through the matching pipeline

1200	648	565	87.2%
Candidates	Ratio survivors	RANSAC inliers	Inlier fraction



Filtered pairs (up to 30 drawn)



Boundary predicted by fitted H

This is a controlled transformed photograph—not two independent camera captures.

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Lab bridge: measure, break, repair

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Measure

Keypoints, matches, inliers, errors and coverage.

Break

Low texture, repetition or extra translation—one at a time.

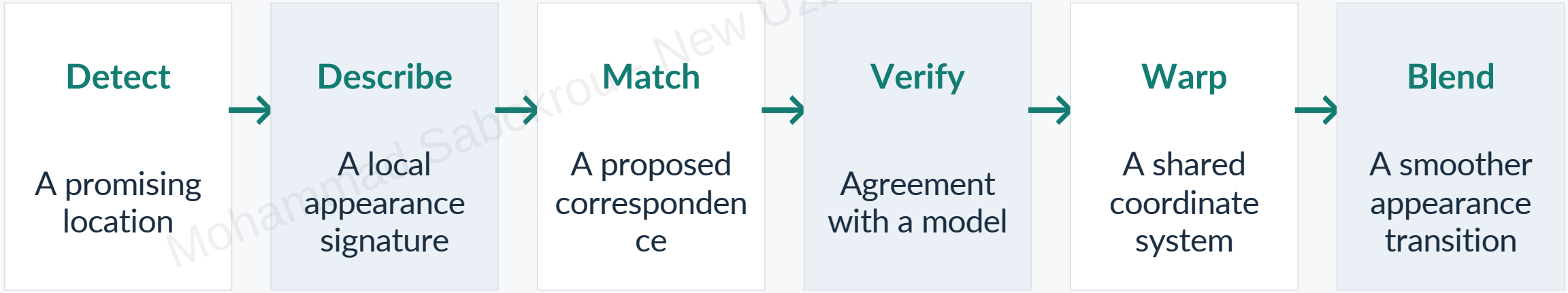
Repair

Choose a change that addresses the observed cause.

Report observations that support your explanation, not only a final image.

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Say what each stage contributes



Appearance evidence

Geometry and construction

A reliable pipeline joins appearance evidence with geometric assumptions.

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Three ideas to take into the lab

01 **A strong feature is not necessarily unique.**

Localisation and identity are different.

02 **A similar descriptor is not necessarily a correct match.**

A ratio threshold is not a probability.

03 **A strong consensus is not necessarily physical truth.**

Inspect assumptions and independent evidence.

Measure what happened. Explain why. Change the right thing.

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Appendix · From patch shifts to gradients

$$E(u, v) = \Sigma\, w(x, y)\, [I(x+u, y+v) - I(x, y)]^2$$

$$I(x+u, y+v) \approx I(x, y) + I_x u + I_y v$$

$$E(u, v) \approx \Sigma\, w(x, y)\, (I_x u + I_y v)^2$$

I: intensity · u, v: small shifts · w: window weight · I_x, I_y: gradients

The approximation is for small shifts and a sufficiently smooth local image.

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Appendix · A complete Harris calculation

Gradients: (2, 0), (0, 2), (2, 2) · all weights = 1

M entries: $\Sigma w\, Ix^2$, $\Sigma w\, IxIy$, $\Sigma w\, Iy^2 \rightarrow 8, 4, 8$

$$M = \begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix}$$

$\det(M) = 8 \times 8 - 4 \times 4 = 48$

$\text{trace}(M) = 8 + 8 = 16$

$k = 0.04$

$$R = 48 - 0.04 \times 16^2 = 37.76$$

All values are illustrative. Thresholds depend on scale and image data.

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Appendix · Why outliers increase RANSAC work

$$N \geq \log(1 - p) / \log(1 - w^s)$$

p: desired all-inlier-sample probability w: assumed inlier fraction s = 4

With p = 0.99	Assumed inliers	Trials (round up)
Easier case	w = 0.50	72
More outliers	w = 0.20	2,876

These trial counts are conditional on the assumptions, not a guarantee of correct geometry.

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Appendix · A safe matching recipe

```
sift = cv.SIFT_create()
k1, d1 = sift.detectAndCompute(gray1, None)
k2, d2 = sift.detectAndCompute(gray2, None)
if d1 is None or d2 is None or len(d2) < 2:
    raise ValueError("Not enough descriptors")

pairs = cv.BFMatcher(cv.NORM_L2).knnMatch(d1, d2, k=2)
good = []
for pair in pairs:
    if len(pair) != 2:
        continue
    m, n = pair
    if n.distance > 0 and m.distance < 0.75 * n.distance:
        good.append(m)
```

No coding is required during the lecture. This is a reference for the lab.

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Appendix · Fit, check and only then warp

```
if len(good) < 4:
    raise ValueError("At least four suitable pairs are needed")
src = np.float32([k1[m.queryIdx].pt for m in good])
dst = np.float32([k2[m.trainIdx].pt for m in good])
src = src.reshape(-1, 1, 2)
dst = dst.reshape(-1, 1, 2)
H, mask = cv.findHomography(src, dst, cv.RANSAC, 3.0)
if H is None or mask is None or not np.isfinite(H).all():
    raise ValueError("A usable homography was not found")
inliers = mask.ravel().astype(bool)
if inliers.sum() < 4:
    raise ValueError("Insufficient geometric support")
# Next: inspect spread, residuals and assumptions before warping.
```

A returned matrix is not sufficient: inspect support, spread, errors and scene assumptions.

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Appendix · A small vocabulary map

Patch	A small neighbourhood of image pixels.	**Keypoint**	A detected position with optional scale and orientation.
Descriptor	A numerical summary of the nearby appearance.	**Correspondence**	A pair intended to represent the same physical point.
Homography	A projective mapping between image coordinates.	**Inlier**	A pair that agrees with a chosen model within a tolerance.
Parallax	Different image displacement for different depths under translation.	**Blending**	Combining appearance in an overlap after alignment.

Use these terms to describe evidence precisely.

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Sources · Foundations and further reading

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A Combined Corner and Edge Detector.
https://www.bmva-archive.org.uk/bmvc/1988/avc-88-023.html

[S] **Lowe (2004), IJCV**
Distinctive Image Features from Scale-Invariant Keypoints.
https://www.cs.ubc.ca/~lowe/papers/ijcv04.pdf

[O] **Rublee et al. (2011), ICCV**
ORB: An Efficient Alternative to SIFT or SURF.
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[R] **Fischler & Bolles (1981), CACM**
Random Sample Consensus, 24(6), 381–395.

[B] **Szeliski (2022), 2nd edition**
Computer Vision: Algorithms and Applications.
https://szeliski.org/Book/

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Sources · OpenCV tutorials and visual credits

[F] **Introduction to Features**
Patch examples and building photograph.
https://docs.opencv.org/4.x/df/d54/tutorial_py_features_meaning.html

[H] **Harris Corner Detection**
Response-map and refinement illustrations.
https://docs.opencv.org/4.x/dc/d0d/tutorial_py_features_harris.html

[SI] **Introduction to SIFT**
Scale-space, extrema and scale-change diagrams.
https://docs.opencv.org/4.x/da/df5/tutorial_py_sift_intro.html

[M / FL] **Feature Matching / FLANN**
Candidate-match images and descriptor metrics.
https://docs.opencv.org/4.x/dc/dc3/tutorial_py_matcher.html

[G / GH] **Homography concepts / object localisation**
Panorama illustration and projected book boundary.
https://docs.opencv.org/4.x/d9/dab/tutorial_homography.html

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